

9/22 Colloquium Notes: Charlotte Ward

Combining the strengths of telescopes for time domain science and cosmology

- LSST @ Rubin $\rightarrow$ Deep Southern Survey
 - Euclid & SPARCS, Grisms, overlap w/ Rubin's, not as deep
 - Roman $\rightarrow$ sparse IR, Deep
 - 30-60% of galaxies overlap in modern Deep surveys
 - Need to anchor ground w/ sparse based imaging
 - Euclid has much wider bands than Roman
- DRP has three modes of joint analysis of increasing difficulty
 - Hardest difficulty is joint detection & pixel modeling
 - o quite the algorithmic problem
 - easy if you simplify $\rightarrow$ low parameters
 - hard if you want to retain morphology $\rightarrow$ every pixel parameter
 - using machine learning priors
 - o What is Jax? $\rightarrow$ clearly a coding language, I've never heard of
 - Cool test to make sure priors are being used sparingly side by side
 - Resampling images makes correlated noise
 - fix: measure correlated noise power spectrum and use it
 - o Slide 27 really shows how powerful this DRP3 method is
 - Trying to incorporate grism spectroscopy into this model
 - This could be powerful $\rightarrow$ accurate phot z's
 - o everything works in observation space
 - o How do you deal w/ continuum? Use Roman filter data $\rightarrow$ large uncertainties
 - o We will detect lensed SM, must be able to model.
- o This algorithm/method of combining data does alot.
 - o ZTF has issues w/ chromatic aberration $\rightarrow$ Refractor? o for nuclear TDEs exist
 - Most CLAGN is northy, hoping to find southern ones
- o Double Peak AGN mentioned

9/22 Colloquium notes continued:

- Double Peaked AGN are "doing their normal thing"
- Need to have Double Peaked AGN model to find more interesting AGN (disk fens, something new, etc.)
- Work with ASKAP to find AGN radio bursts
- Radio emission comes later than a Visual for AGN flares
- NGC 1566 is a changing look AGN $\rightarrow$ does SPT see this?

Questions: The survey combination pipeline DDP 3 & Scalet 2 take data from telescopes combines it and then does data analysis on the combined data. This is similar to interferometry where the difference is data is combined ~~as~~ as counts because there is no heterodyne detection. Do people who work on Interferometry have insight that could improve this model? Hogg this group learned anything that can be applied to these pipelines?

- With the Orion, if the pipeline bundles the data successfully, would this be able to determine spec z for every source?